Course notes

Meaning of a limit

$f(x) \to \ell$ when $x \to +\infty$ means that $f(x)$ can be made as close as desired to ℓ for all x sufficiently large. For a limit at a, consider x of the domain, close to a and distinct from a. Left and right limits are written $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$. When the domain has points on both sides of a, a two-sided limit exists if and only if the two one-sided limits exist and are equal. At an endpoint of the domain, we study only the limit accessible side.

Graphical interpretation

If $f(x) \to \ell \in \mathbb{R}$ when $x \to +\infty$ or $-\infty$, the line $y = \ell$ is a horizontal asymptote at this direction. If $f(x) \to \pm\infty$ when $x \to a$ on one side, the line $x = a$ is an asymptote vertical.

A curve can cross a horizontal asymptote. The limit does not necessarily give the position of the curve relative to the asymptote: study the sign of $f(x) - \ell$.

References and operations

At $+\infty$, $x^m \to +\infty$ for $m \geq 1$, $\frac{1}{x^m} \to 0$, $\sqrt{x} \to +\infty$. In $-\infty$, the sign of x^m depends on the parity of m. The rules for sum, product, quotient and comparison are those for sequences, with the same forms undetermined.

Example. For $f(x) = \frac{2x+1}{x-3}$, write $f(x) = 2 + \frac{7}{x-3}$. The limits at $\pm\infty$ are equal to 2. At 3^-, $f \to -\infty$; at 3^+, $f \to +\infty$. The asymptotes are $x = 3$ and $y = 2$.

Conjugate. For $x > 0$, $\sqrt{x^2+4x} - x = \frac{4}{\sqrt{1+\frac{4}{x}}+1} \to 2$. For $x < 0$, be careful: $\sqrt{x^2} = |x| = -x$.

Exponential and comparative growth

$$\lim_{x\to+\infty} e^x = +\infty, \quad \lim_{x\to-\infty} e^x = 0, \quad \lim_{x\to+\infty} \frac{e^x}{x^m} = +\infty \quad (m \in \mathbb{N}).$$

We deduce $x^m e^{-x} \to 0$ at $+\infty$.

Proof of the limits of the exponential. The study of $e^x - x - 1$ gives $e^x \geq 1 + x$. By comparison, $e^x \to +\infty$ in $+\infty$. Setting $t = -x \to +\infty$, $e^x = \frac{1}{e^t} \to 0$.

Proof of a comparative growth result. For every integer $k \geq 0$ and $x \geq 0$, $e^x \geq \frac{x^k}{k!}$. For $k = 0$, $e^x \geq 1$. If the inequality holds at index k, the derivative of $e^x - \frac{x^{k+1}}{(k+1)!}$ is $e^x - \frac{x^k}{k!} \geq 0$; its value at 0 is positive. The inductive step follows. By choosing $k = m + 1$, $\frac{e^x}{x^m} \geq \frac{x}{(m+1)!} \to +\infty$.

Choose a transformation

Quotient of polynomials: factor out the highest powers. Difference of square roots: quantity conjugate. Quotient near a zero: factorise and study the signs. Exponential versus power: comparative growth, after a change of variable if necessary.

Comparison and composition: conditions of use

If $g(x) \leq f(x) \leq h(x)$ near the point or infinity studied, and if g, h have the same real limit ℓ, then $f(x) \to \ell$ (squeeze theorem). If $g \leq f$ and $g \to +\infty$, then $f \to +\infty$. For a limit $-\infty$, we uses an upper bound by a function tending to $-\infty$. If $u(x) \to a$ and v is continuous at a, then $v(u(x)) \to v(a)$, when the composite is defined. For example $\sqrt{1+\frac{1}{x}} \to 1$ at $+\infty$. For a limit infinite of the outer function, check from which side $u(x)$ approaches the bound: $\frac{1}{u(x)}$ does not have the same behaviour according to the sign of $u(x)$.

Read a limit, choose a transformation

3.1 The approach side is part of the question

First use: S23.

Worked example · A vertical asymptote

For $f(x) = 1 + \frac{1}{x-2}$, the domain is $\mathbb{R} \setminus \{2\}$. When $x \to 2^-$ the denominator is negative and tends to 0, so $f(x) \to -\infty$. When $x \to 2^+$, $f(x) \to +\infty$. The line $x = 2$ is an asymptote vertical; no value $f(2)$ exists. At infinity, $f(x) \to 1$.

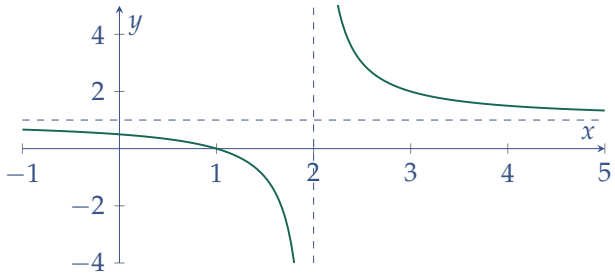


Figure 3.1. Find the two one-sided limits on the drawing: the branches are not joined.

A1 · Immediate application Determining the limits of $g(x) = 3 - \frac{2}{x+1}$ in -1^-, -1^+, $-\infty$ and $+\infty$. Give the asymptotes.

3.2 An indeterminate form calls for a calculation

First use: S24.

Worked example · The sign hidden in a square root

For $x < 0$, $\sqrt{x^2} = |x| = -x$. Thus, in $-\infty$,

$$\frac{\sqrt{x^2+1}}{x} = \frac{|x|\sqrt{1+1/x^2}}{x} = -\sqrt{1+1/x^2} \longrightarrow -1.$$

At $+\infty$, the same expression tends to 1. Omitting the absolute value would produce a limit positive for an expression that is actually negative.

A2 · Fading guidance Calculate in $+\infty$ the limits of $\frac{2x^2+1}{x^2-x}$, $\sqrt{x^2+4x} - x$ and $x\left(\sqrt{1+1/x} - 1\right)$. For the first, we suggest dividing by x^2 ; choose freely for the others.

A3 · Construct examples Construct three quotients whose limit has the form $0/0$ at 0, whose limits are respectively 0, 1 and $+\infty$. State the side of approach if necessary.

3.3 Comparative growth and change of variable

First use: S26; logarithm after S49.

Property · Exponential and powers · results from the lesson

For every integer $m \geq 1$, $e^x/x^m \to +\infty$ and $x^m e^{-x} \to 0$ at $+\infty$. To transfer these results, identify the new variable and its limit.

Worked example · Transferring a result

When $x \to -\infty$, set $t = -2x$ gives $t \to +\infty$ and $x^2e^{2x} = \frac{1}{4}t^2e^{-t} \to 0$. After S49, the substitution $t = 1/x$ in $x \ln x$ for $x \to 0^+$ gives $-(\ln t)/t \to 0$. The logarithm does not occur not before studying it.

A4 · Deferred application After S26, calculate $\lim_{x\to+\infty}(x^2+3x)e^{-x}$ and $\lim_{x\to-\infty} xe^x$. Say why formally replace x by infinity is not a justification.

3.4 An asymptote may be crossed

First use: S24.

Definition · Horizontal Asymptote

If $f(x) - \ell \to 0$ at one of the infinities, $y = \ell$ is an asymptote at this infinity. This definition requires a vertical distance that tends to 0; it does not rule out any intersection at a finite distance.

Worked example · Two behaviours

On $\mathbb{R}$, $p(x) = 1 + 1/(1+x^2)$ remains strictly above $y = 1$. The function $q(x) = 1 + x/(1+x^2)$ crosses this same line at $x = 0$, while tending to 1 at both infinities. For q, the sign of $q(x) - 1$ is that of x.

A5 · Refute and explain A student says that the curve of q cannot cross its asymptote. Give the coordinates of the point of intersection and the signs on either side. Is this consistent with the two limits?

A6 · Choice of strategy Determine a, b so that $h(x) = \frac{ax+b}{x-1}$ has the horizontal asymptote $y = 2$ and satisfies $h(0) = 3$. Then study the one-sided limits at 1 and the position of the curve relative to $y = 2$.

One minute, spoken aloud. For each of the expressions in the "Methods" exercise, first state which prevents concluding using the operations on limits, then explain the role of the transformation.

Delayed review · S49 then S81. Redo the change of variable with $\ln x/x$; sort again the limits combining logarithms, exponentials and roots.

Investigation

A curve close to a line - 30 min.

Consider the function f defined on $\mathbb{R} \setminus \{1\}$ by $f(x) = \frac{3x-2}{x-1}$. Let $\mathcal{C}_f$ its curve in a coordinate system of the plane.

1. Show that, for every $x \neq 1$, $f(x) = 3 + \frac{1}{x-1}$.
2. Determine the limits of f in $-\infty$, at 1^{-}, at 1^{+} and at $+\infty$.
3. Study the sign of $f(x) - 3$, then specify the position of $\mathcal{C}_f$ with respect to the line with equation $y = 3$.
4. Determine a real number $A > 1$ such that, for every $x > A$, $|f(x) - 3| < 0{,}01$.
5. The curve $\mathcal{C}_f$ intersect its horizontal asymptote? Justify and explain why a window an overly wide graph can give a misleading impression.